Breast-gut connection: origin of chenodeoxycholic acid in breast cyst fluid.

The notion that a breast-gut connection might modulate the microenvironment of breast tissue was supported by the finding that breast cyst fluid contains bile acids that are characteristically found in the intestines. To establish that the gut, rather than circulating steroid precursors, is the source of bile acids in breast cyst fluid, we gave two patients deuterium-labelled chenodeoxycholic acid (three 200 mg doses by mouth), starting 9 days before aspiration of breast cysts. The chenodeoxycholic acid concentration of seven samples of aspirated cyst fluid ranged from 42 to 94 mumol/L. The corresponding serum concentrations of chenodeoxycholic acid on the same day were 0.8 and 2.9 mumol/L, of which the labelled compound comprised 13.0% (0.38 mumol/L) and 28.2% (0.23 mumol/L). The deuterated chenodeoxycholic acid concentrations in cyst fluid were 0.79 and 1.26 mumol/L in two samples from patient 1 and 3.22 mumol/L in patient 2; these values are equivalent to 11-17% of the serum concentrations [corrected]. This study shows that intestinal bile acids rapidly gain access to cyst fluid. Further studies should investigate the mechanisms that govern the exchange processes and the maintenance of the high cyst fluid to plasma concentration gradients, and the biological half-lives of individual constituents.